A3-PS018-TC381

HAPS Link-Quality Prediction Using a Documented Channel Model

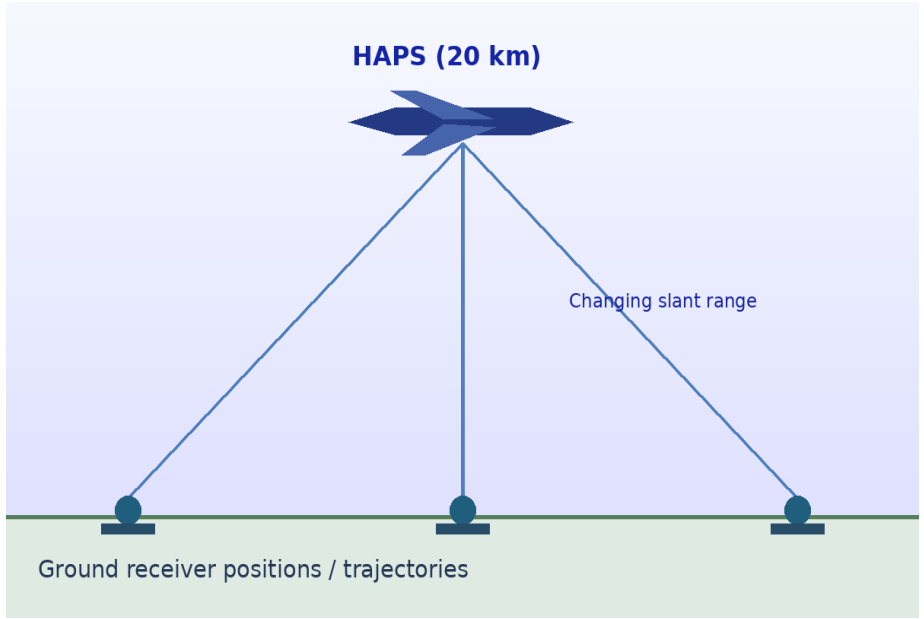
Team Strato | AIORI Hackathon 3

Ramita K · ramita11k@gmail.com
Mokshapriya T A · ttamokshapriya@gmail.com
Mentor: Trupti Tagare · truptitagare-ece@dayanandasagar.edu

October 14, 2026

Problem Description

- **HAPS** supports high-altitude wireless links.
- **Received power** varies with geometry and signal loss.
- A **documented channel model** enables reproducible estimates.
- Goal: predict **received power** and compare with **analytical** values.



Conceptual **HAPS**-to-ground link geometry

Background

- Fixed HAPS altitude: **20 km**; one selected operating frequency.
- Distance, elevation angle and **atmospheric** absorption influence propagation.
- A **link budget** converts propagation loss into received signal power.
- **Regression** can approximate modeled received power on unseen trajectories.

$$P_r\ (\text{dBm}) = P_t\ (\text{dBm}) + G_t\ (\text{dBi}) + G_r\ (\text{dBi}) - L_{\text{total}}\ (\text{dB})$$

Literature & RFC Review

ITU-R P.525	Free-space path loss for the fixed HAPS geometry
ITU-R P.676	Gaseous attenuation along an Earth–space path
ITU-R P.835	Reference atmospheric profiles for scenario definition

Model applicability and profile assumptions will be verified during implementation.

Proposed Solution

- Set one HAPS altitude, frequency and documented link-budget assumptions.
- Generate **reproducible synthetic data** across ground-receiver trajectories.
- Evaluate **two defined environmental scenarios** using compatible models.
- Train a regression model to predict **received power (dBm)**.
- Compare against analytical calculations on **held-out trajectories** using **MAE**.

Optimization Proposed

Objective: test whether regression offers efficient approximation of analytical link calculations.

Evaluation aspect	Analytical reference	Proposed ML-assisted approach
Output	Link-budget calculation	Predicted received power
Accuracy	Reference target	Held-out trajectory MAE
Computation	Repeated model evaluation	Measure inference time
Sensitivity	Vary propagation inputs	Check prediction response

Solution Architecture & Design

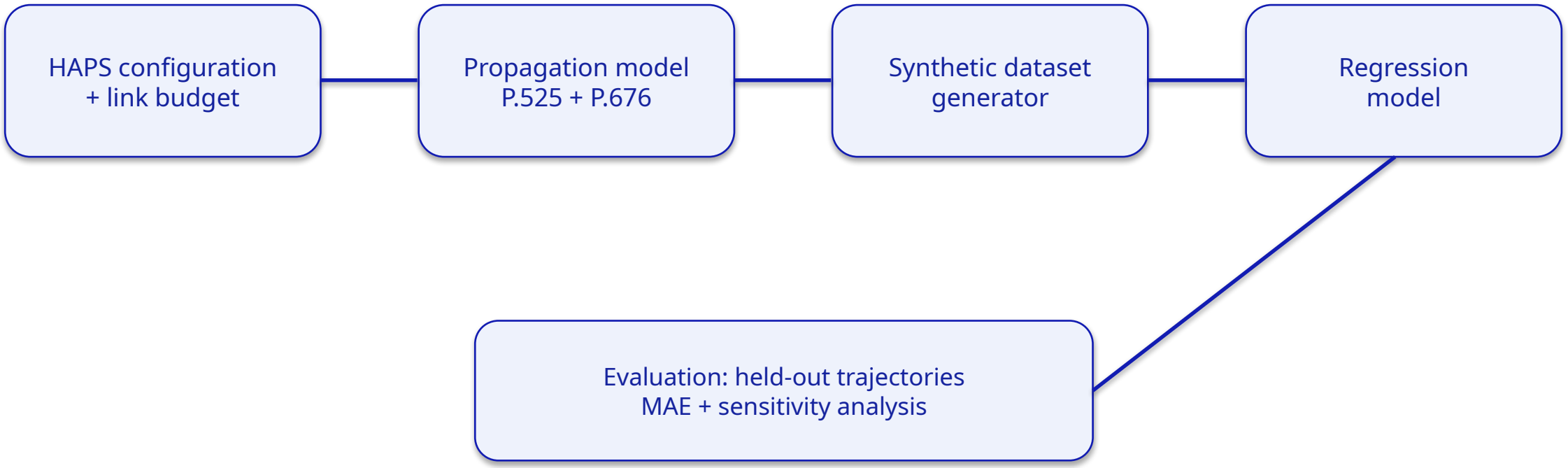


Figure: Proposed system architecture (conceptual)

Data Flow Diagram

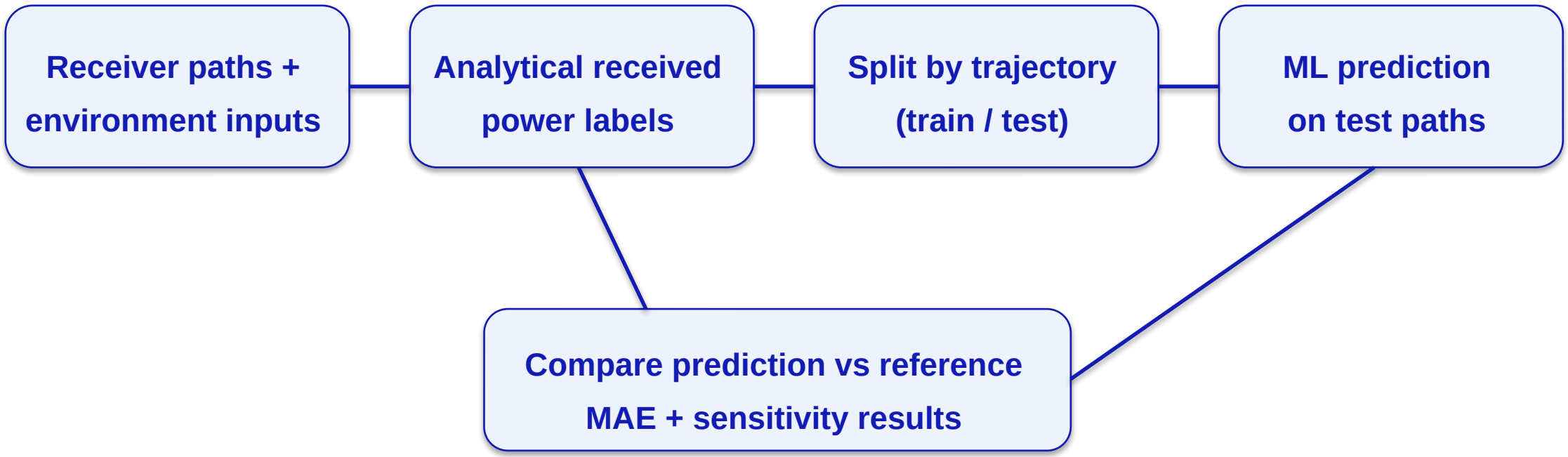


Figure: Proposed data flow; test trajectories are not used for training

Deployment Timeline

1. Document assumptions and select scenario profiles
2. Implement analytical link budget and dataset generator
3. Generate and split receiver trajectories reproducibly
4. Train baseline and Random Forest regression models
5. Evaluate MAE, sensitivity and computational cost
6. Prepare technical report, demonstration and GitHub files

Milestones are proposed; exact dates will follow the official hackathon schedule.

Bibliography

- ITU-R P.525 — Calculation of free-space attenuation.
- ITU-R P.676 — Attenuation by atmospheric gases and related effects.
- ITU-R P.835 — Reference standard atmospheres.
- AIORI-3 — PS-018: HAPS Link-Quality Prediction Using a Documented Channel Model.

Thank You for Your Attention!

Team STRATO · A3-PS018-TC381